

Dart Programming Fundamentals

Mobile App Development with Flutter



What is Dart?



Dart is a modern, object-oriented programming language developed by Google specifically for building beautiful, high-performance mobile, web, and desktop applications.

As the foundation of Flutter, Dart enables developers to create native apps for iOS, Android, and beyond using a single codebase. Its fast execution and developer-friendly syntax make it perfect for UI-intensive applications.

Why Learn Dart?



Flutter Foundation

Essential language for Flutter development, the fastest-growing cross-platform framework



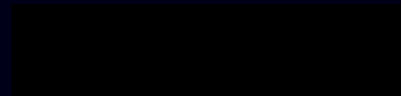
Beginner-Friendly

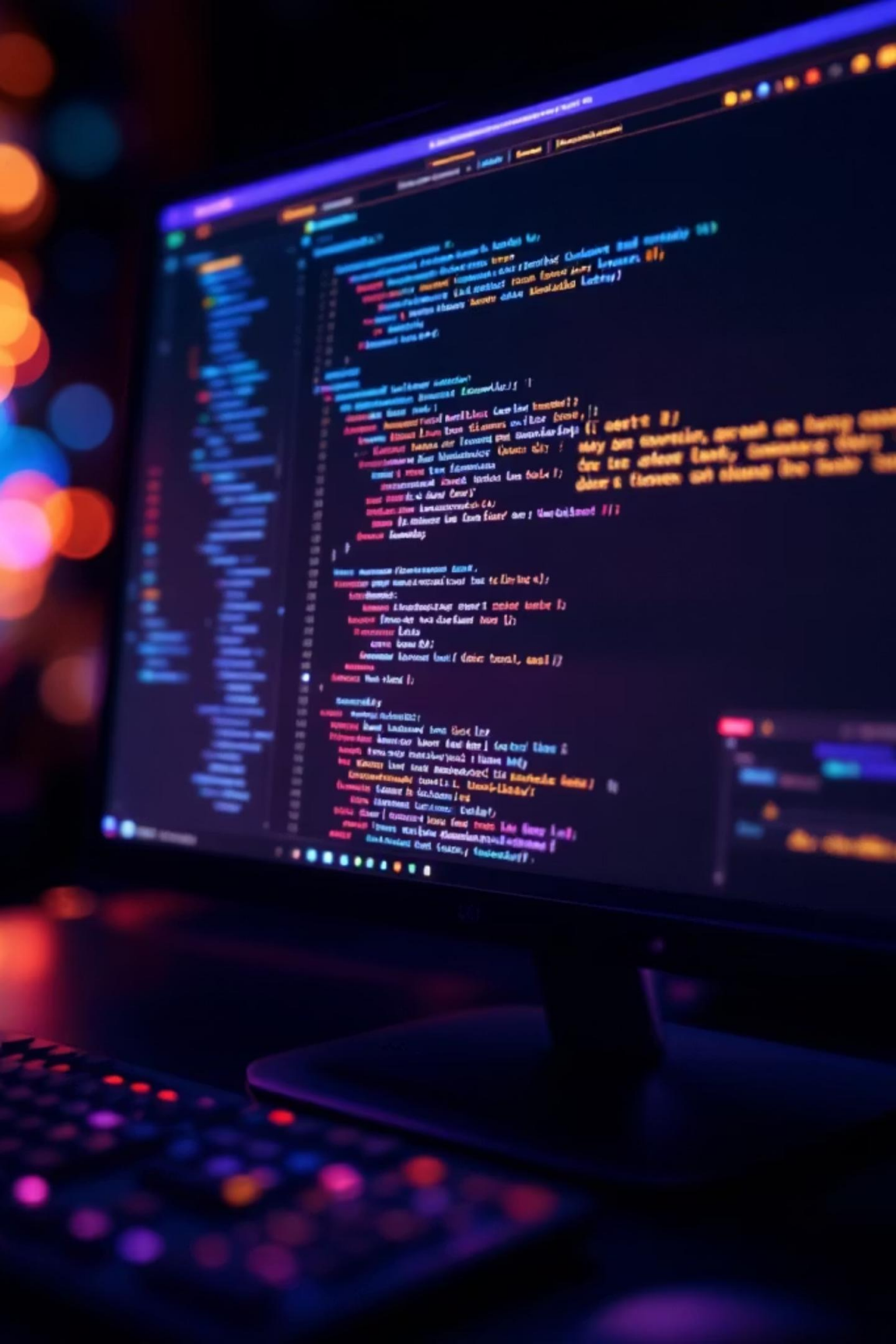
Easy to learn with syntax similar to Java and JavaScript



Modern Features

Null safety, async programming, and powerful type system





Getting Started: Development Environment

DartPad

Online code editor by Dart team

- No installation required
- Perfect for learning
- Instant feedback

What You Need

Just a web browser and internet connection

- Access at dartpad.dev
- Free to use
- Save your work

Structure of a Dart Program

Entry Point

Every Dart program starts with the `main()` function. This is where execution begins when you run your program.

Output Function

The `print()` function displays text to the console, helping you see program output and debug issues.

```
void main() {  
  print("Hello, Dart!");  
}
```

Variables and Data Types

Variables store data in your program. Dart supports several core data types that help you work with different kinds of information.

int

Whole numbers like 42, -7, or
1000

double

Numbers with decimals like 3.14
or -0.5

String

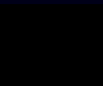
Text enclosed in quotes like
"Hello"

bool

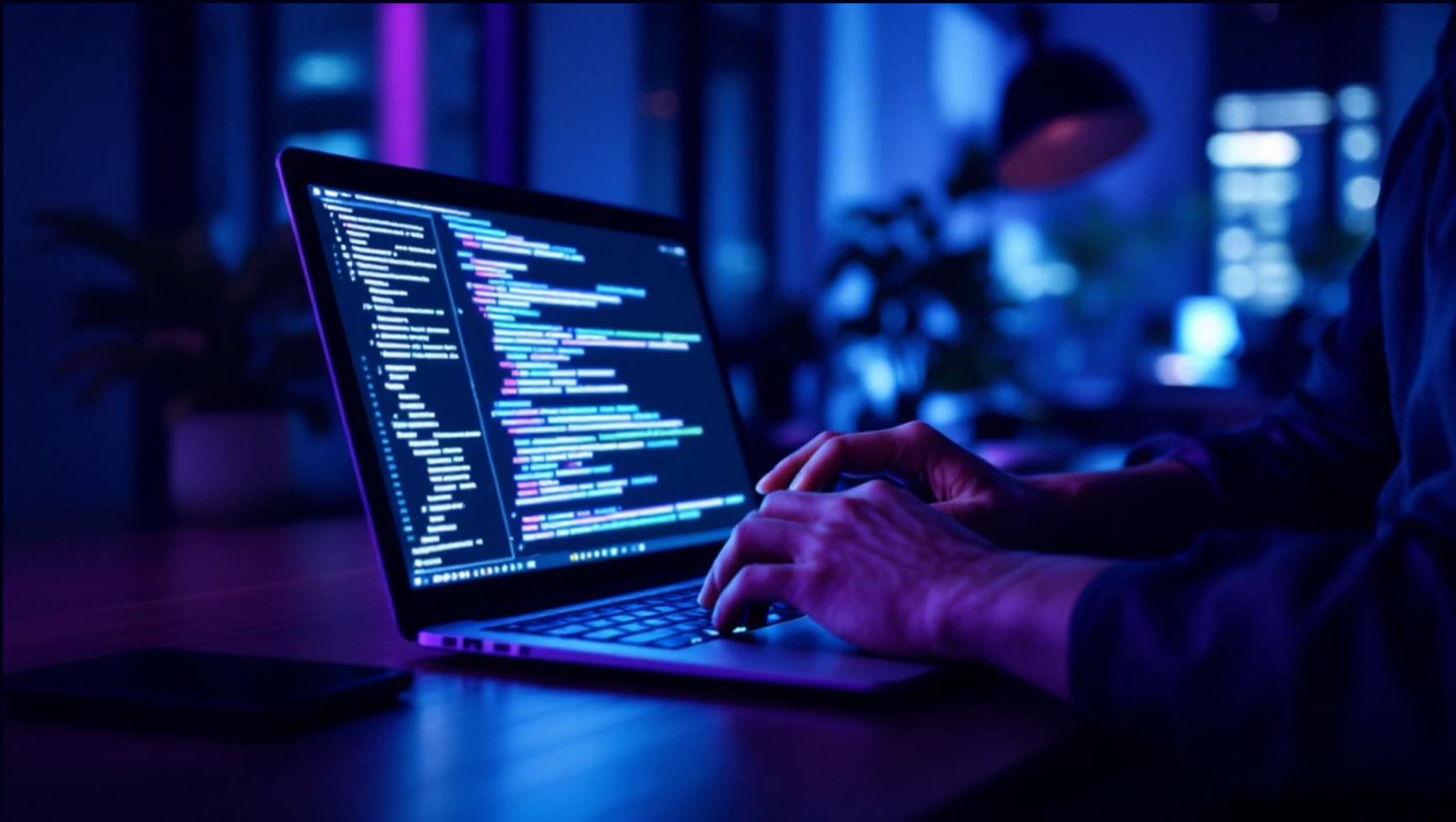
True or false values

var

Let Dart decide the type
automatically



Declaring Variables



Explicit Typing

Specify the data type when declaring variables for clarity and safety.

```
String name = "Ali";  
int age = 22;  
double height = 5.9;  
bool isStudent = true;
```



Type Inference with var

Dart can automatically detect the type of your variable based on the value you assign. This makes your code cleaner and easier to write.

```
var city = "Karachi";           // String
var score = 95;                 // int
var price = 29.99;              // double
var isActive = false;          // bool
```

The **var** keyword tells Dart to figure out the type for you, reducing repetitive code while maintaining type safety.

Conditional Statements

Control the flow of your program using conditions. The **if-else** statement lets you execute different code based on whether a condition is true or false.

```
var marks = 75;

if (marks >= 50) {
  print("Pass");
} else {
  print("Fail");
}
```

1

Condition

Check if marks is 50 or higher

2

True Path

Print "Pass" if condition is met

3

False Path

Print "Fail" otherwise



Practice Exercise

01

Open DartPad

Visit dartpad.dev in your browser

02

Create Variables

Declare a name (String) and age (int) variable

03

Add Condition

Write an if-else statement to check if `age >= 18`

04

Test Your Code

Run the program and verify the output

